Sangjun (Samuel) Ko

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Work Experience

Kimberly-Clark — North America Family Care R&D Neenah, Wisconsin
Statistician (contract assignment through Staffmark Group) June 2025 – Present

- Serve as statistical lead for Test Method Validation (TMV) and measurement systems across Family Care R&D; designed balanced factorial equivalence studies ($N = 96$) that detected a persistent 0.7–1.3 mg negative bias on candidate hardware, halting premature tool adoption and driving supplier re-machining.
- Design multi-factor screening DOEs (including 96-run cart comparison and 48-run fluid insult rate optimizations) to isolate critical process and test parameters, correlating accelerated benchtop testing with consumer home-use data to screen prototypes in weeks.
- Formulate Test Method Characterization (TMC) and Gage R&R protocols for destructive fastening tab-tear strength; decompose single-pull variance components and establish statistical guardbands against process defect thresholds.
- Engineered an automated Python time-series data reduction pipeline for transient sensor curves; modeled moisture dissipation dynamics to establish an empirical 14–15 minute plateau rule, eliminating operator timing variation.
- Conduct retrospective validation on multi-variant adhesive weight data; separate operator measurement variability from true manufacturing dispersion using statistical tolerance intervals to establish variant-specific release specifications.
- Rebuilt core product-performance modeling engine from legacy SAS into a containerized Python/Flask web app, resolving split-plot error structures and reducing report generation time from days to minutes for 50+ researchers.

Applied Technologies for Learning in Arts & Sciences Champaign, Illinois
AI/ML Intern Team Leader January 2024 – December 2024

- Led five interns and coordinated deliverables with three managers while developing and evaluating Python predictive machine-learning models across datasets with 100,000+ records.

University of Illinois Urbana-Champaign Champaign, Illinois
Undergraduate Course Assistant — Data Science and Professional Statistics August 2023 – May 2025

- Mentored 300+ students in data analysis, experimental design, and statistical consulting; managed Git-based instructional content for a learning platform serving 1,000+ students.

HappySneeze Remote
Market Research Intern August 2024 – September 2024

- Created an interview guide, conducted 10+ consumer interviews on healthcare needs, and synthesized findings into actionable product and design recommendations.

Education

University of Illinois Urbana-Champaign Champaign, Illinois
B.S. Statistics, Mathematics Minor; GPA: 3.93/4.00 May 2025

University of Michigan–Shanghai Jiao Tong University Joint Institute Shanghai, China
Mechanical Engineering coursework; transferred to UIUC; First Class Scholarship 2019 – 2020

Technical Skills

Process & Quality:	Design of Experiments (DOE), Test Method Validation (TMV), Gauge R&R/MSA, tolerance intervals, guardbanding, equipment equivalence, root-cause analysis, variance components
Programming & Tools:	Python, JMP, Flask, Excel/VBA, R, SQL, Git, JavaScript, HTML/CSS; Pandas, NumPy, SciPy, statsmodels, scikit-learn
Languages:	Korean and Spanish (native); English (fluent)

Service

Volunteer, Appleton Public Library August 2025 – Present

- Tutor middle- and high-school students in math weekly; previously supported "Game On" teen programs.